

Symmetric lens

MathlibAnnex.symmetricLens

def

Names the intersection of two closed balls with a common radius.

Statement

Let E be a normed additive commutative group, let $x, y \in E$, and let $r \in \mathbb{R}$. The symmetric lens with foci x, y and radius r is the set $\Lambda_r(x, y)$ defined below.

Definition

Write $\overline{B}(x, r) = \{z \in E : \|z - x\| \leq r\}$. Then

$$\Lambda_r(x, y) = \overline{B}(x, r) \cap \overline{B}(y, r) = \{z \in E : \|z - x\| \leq r \text{ and } \|z - y\| \leq r\}.$$

Thus membership imposes both closed distance bounds with the same radius.

Assumptions

Only the normed additive commutative group E , two points x, y , and a real number r are required. The foci may coincide, and r need not be positive.

Conclusion

The output is a subset of E , symmetric in its two foci: $\Lambda_r(x, y) = \Lambda_r(y, x)$. If $r < 0$, it is empty because norms are nonnegative. No midpoint or reflection assertion is part of this definition.

Main citations

- Exact definition and context

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
-- The intersection of two closed balls with the same radius. -/
def symmetricLens (x y : E) (r : ℝ) : Set E :=
  closedBall x r ∩ closedBall y r
```

In the source	Mathematical meaning
$(x\ y : E)\ (r : \mathbb{R})$	The two points x, y of the normed additive commutative group E , and their common real radius r . No real scalar multiplication or positive-radius hypothesis is required.
$: \text{Set } E$	The output $\Lambda_r(x, y)$ is a set whose elements are points $z \in E$.
$\text{closedBall } x\ r\ \cap\ \text{closedBall } y\ r$	The complete right-hand side: $z \in \Lambda_r(x, y)$ exactly when $\ z - x\ \leq r$ and $\ z - y\ \leq r$.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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